

E.L. Hynes PhD

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The Hague, The Netherlands

Artist in Residence Varna, Water Tower Arts Festival 2021

Nederlands Octrooibureau B.V, The Hague, The Netherlands: Jan 2020 - current
UK Patent Attorney / European Trainee Attorney

Drafting and prosecuting patent applications in the UK, Netherlands, Europe and internationally.
Mechanical and electrical engineering, information and communications technology, chemistry, and materials inventions.

Part of medical devices and quantum technology specialist interest groups.
Published articles about coronavirus patent commons and patenting challenges for quantum technologies.

Uruqhart-Dykes and Lord, Milton Keynes, UK: March 2018- January 2020

Trainee patent attorney

Drafting and prosecuting patent applications in the UK, Europe and internationally.
Introduced a new sector on the company website for green technology including five blog articles, also published on Lexology in the top twenty read articles by the firm.

Bournemouth University January 2019-July 2019

Postgraduate Certificate in Intellectual Property

Awarded with Merit.

Swansea University: January 2015-June 2018

PhD Researcher

Testing polymer theory for thin-film polymer-fullerene bilayers using neutron reflectivity, grazing incidence x-ray diffraction, atomic force microscopy and polarised optical microscopy. Research directed at improving materials in organic photovoltaic devices.

Discovered film thickness dependence of morphology of fullerene crystals and molecular weight dependence of interfacial width in model system.

Awarded Best Doctorate in the Systems and Processes Engineering Centre 2018.

Presented frequently.

Research published in Nature Communications Physics and Soft Matter journals

- ISIS Rutherford Appleton Laboratory Neutron Training School: July 2015
Doctoral training school in neutron scattering
- Loughborough University: March 2016 'Solutions in the Spring' (Institute of Physics)
Doctoral training school on soft matter.

Swansea University, UK: 2015-2018

Lab demonstrator for MSc

Atomic force microscopy, scanning tunnelling microscopy and preparation of polymer bilayers.

Advocated health and safety protocol.

Taught the students the underlying theory and applications.

Swansea Sixth Form College: 2016-2017

Teacher

GCSE and A Level Maths and physics as well as Qualified Teacher Status Numeracy Skills.

MA Art and Science student Central Saint Martin's 2013-2014

Certificate in Bronze Sculpture NCAD 2012-2013

University College Dublin Sep 2008-2012

BSc First Class Hons Theoretical Physics (GPA: 3.83)

Advanced quantum mechanics, electromagnetism, thermodynamics, applied mathematics, fluid mechanics, computational science, electronics and programming (java, C, C++ and MATLAB)

Performed a study of the interrelationship of special relativity and electromagnetism.

Competed in both Maths and Physics intervarsity competitions several times.

Three scholarships for academic performance in 2008 and 2010.

- University College Dublin /Trinity College Dublin Maths Summer School Jun 2011
- Universitas 21 International Summer School in Conflict Resolution Aug 2011

University College Dublin Physics Department: Jun- Aug 2010

Summer undergraduate researcher

Simulation of micro-robotic propulsion in a high viscosity environment

Presented my work to the faculty.

Conferences

Physical Aspects of Polymer Science, Swansea University; September 2017

GRAPOL IOP conference, London 2017

Hybrid and Organic Photovoltaics; Swansea University; June 2016

PhD Users Meeting; ISIS Abingdon; October 2015

Physical Aspects of Polymer Science; Manchester University; September 2015

Printed Photovoltaics- From materials to manufacture; Swansea; April 2015

Soft Matter Journal Club Swansea University; May 2016 and 2017

First Author Publications

Interfacial width and phase equilibrium in polymer-fullerene thin-films; Nature Communications Physics 2018 ([link here](#))

Liquid-liquid equilibrium in polymer-fullerene mixtures;
an in-situ neutron reflectivity study; Soft Matter 2020 ([link here](#))

Mixing and Interfacial Composition in Polymer/Fullerene
Thin Films Nature; PhD Thesis 2019 ([link here](#))

Lexology ([link here](#))

World Intellectual Property Review- Patenting Quantum Technology ([link here](#))

Lessons for COVID-19 from the Eco-Commons ([link here](#))

References

Prof Anthony Higgins- a.m.higgins@swansea.ac.uk

Prof Vladimir Lobaskin- lobaskin@ucd.ie